

Endomorphisms

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1 Introduction

This should (eventually) read like a taxonomy of the elements of the endomorphism of some set. If cardinality arguments appear and look weird, assume the underlying set is finite.

2 Investigation

So the first thing I want to lock is notation. For a given set S , the set of functions $S \rightarrow S$ will be written as E_S . It is not pretty but it is easy to type. If we ever need to think about what a given $f \in E_S$ does to some $s \in S$, we shall not write it as $f(s)$ but instead we will compose the constant function $f_s(x) = s$ with f . Immediately we will abuse notation and say that s denotes both the element of S and the constant function.

Another thing we will do is write function products as follows

$$fg = g \circ f. \tag{1}$$

This of course generalizes to longer products like $fhghfghfhg$, which are associative since composition is associative. Then, instead of evaluating functions (if we need to work with element traces/paths) we will merely write

$$s_0 f \tag{2}$$

where s_0 is the constant function $s_0(x) = s_0 \in S$ and f is some element of E_S . We will name set elements after the set, so if S is a set the symbols $s_0, s_1, \dots, s_n, \dots$ are both constant functions and elements of S . For a set C , we reserve $c_0, \dots$ and so on.

Finally, when speaking about cardinalities, it probably safer to assume finite sets, so we have the common arithmetic available. I know wonder what would take for set theory to allow for arbitrary cardinalities. What is a set of size 2.5?

3 Synthesis

Definition 3.1: Let S be a set. The set of functions $f : S \rightarrow S$ will be called the **endomorphism set**, or the **function set**, and will be written as E_S . $\bigcirc$

Definition 3.2 (Function Product): Let S be a set and E_S it's function set. The product of two functions $f, g \in E_S$ is defined as their composition:

$$fg = g \circ f. \quad (3) \quad \bigcirc$$

Note 3.3: The previous definition is done in such a way as that reading left-to-right also tells us the order of application of the functions. $\triangle$

Proposition 3.4: The set of endomorphisms of some set S with the product of 3.2 forms a monoid.

Proof: Because the product is defined from the composition, it inherits the associativity and the neutral element $i(x) = x$, the identity function.

Nevertheless, we show some calculations¹. Let $f, g, h \in E_S$ and consider $f(gh)$. By definition,

$$f(gh) = (gh) \circ f \quad (4)$$

which yields

$$(gh) \circ f = (h \circ g) \circ f \quad (5)$$

which we associate freely, since it's in composition form, to

$$(h \circ g) \circ f = h \circ (g \circ f), \quad (6)$$

giving us

$$h \circ (g \circ f) = h \circ (fg) = (fg)h \quad (7)$$

as expected.

Consider the function $i(x) = x$ in E_S . By: definition, composition property, composition property, and definition, we have $fi = i \circ f = f = f \circ i = if$. $\blacksquare$

Note 3.6: We will not justify every equality like in the last proposition. Do not worry.

The next proposition serves to establish our notation. $\triangle$

Proposition 3.7 (No More Parentheses): Let S be a set, E_S it's function set and $\bar{s}(x) = s \in S$ be a constant function. For every function $f \in E_S$ we have

$$\bar{s}f = f(s). \quad (8)$$

¹These are calculations I actually understand, so I might as well show them.

Proof: We just apply our definition. Since $\bar{s}(x) = s$, when we compute $\bar{s}f$ we obtain

$$\bar{s}f = f \circ \bar{s} \quad (9)$$

which applied to some $x \in S$ yields

$$(f \circ \bar{s})(x) = f(\bar{s}(x)) = f(s) \quad (10)$$

as expected. ■

Note 3.9: The previous argument exists for the sole purpose of freeing us from having to write $f(s)$ for $s \in S$ should we ever need to think about where an element goes. We will also abuse the notation and consider $s \in S$ both as an element and as it's constant function $s(x) = s$. △

Proposition 3.10 (Rightmost Zeroes): For a given set S and it's endomorphisms, any constant function $s \in E_S$ is such that $fs = s$ for all $f \in E_S$.

Proof: If we expand the product into the composition

$$fs = s \circ f \quad (11)$$

and remember s is a constant function, then

$$s \circ f = s \quad (12)$$

and then $fs = s$ as expected. ■

Note 3.12: Two things we ought to point out now:

1. We will write the identity function $f(x) = x$ as $e \in E_S$.
2. The last two propositions (and others to come) remain true if instead of a single function we compose with a finite product of functions

$$P = \prod_{i=1}^{n \in \mathbb{N}} f_i. \quad (13)$$

In other words, sP is still equal to evaluating $P(s)$ and $Ps = s$. △

4 Not Conclusions

Next step is actually doing some taxonomy on E_S . We have bijections, constant functions. I know the literature calls the elements of the function set transformations of the set but I will instead call them k -classifications based on the size of their image, because I will actually use the k in an equation.

I am writing this alone. Why do I use plural language persons? We don't know.